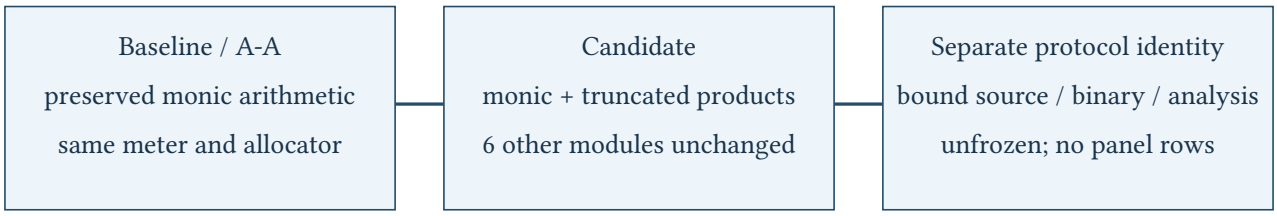
Isolated truncated-product accounting

Prime-field native source · 2026-11-03

The new matched runner uses monic arithmetic as baseline and A/A, with truncated monic multiplication as the candidate. Six non-polynomial modules are byte-identical between the arms. This isolates the low-product change from the preceding monic specialization.



Checked instrument

Seven generated modules reverse exactly to original source bytes. Existing helper and schoolbook-update hooks cover the low-product leaves; two new hooks cover shifted assembly additions. All 29 prebuild source/dependency hashes pass.

Optimized production and test binaries compile. Two selected methods pass, with 85 inherited methods filtered. The triangular leaf contract matches $n(n+1)/2$ products and additions at $n=1,2,4,8,12$. All eight complete instrumented recovery receipts agree with monic. These checks establish instrument behavior, not a cost-reduction result.

Separate comparison identity

The new draft binds its source commit, production binary and unchanged strict analyzer. Canonical validation reports zero findings. The existing monic panel and original 4563-case cohort are unchanged. Preflight, remaining budget and authoritative EKS admission remain before protocol freeze and data collection; no panel or cloud job ran.

Evidence

See source-lock.json, instrumentation.json, receipt.json, raw build/test logs and comparison-protocol.draft.json. Prior method context is the low-product derivation and FLINT truncated-product documentation.